

T Karthik Reddy

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SUMMARY

Software Engineer focused on distributed systems, AI infrastructure, and scalable backend services. Experienced building production GenAI platforms and high-performance concurrent systems using C++, Rust, Java, and Python.

EDUCATION

Mahindra Ecole University

BTech – Computer Science Engineering, CGPA: 7.59/10

Nov 2022 – Jul 2026

Hyderabad, India

- **Coursework:** Operating Systems, Distributed Systems, Computer Networks, DBMS, Algorithms, Compilers, Machine Learning

PROFESSIONAL EXPERIENCE

Techolution

Associate AI Engineer

Mar 2026 – Present

Hyderabad, India

- Implemented a flagship **healthcare BPA platform**, compressing a 6-month manual document process into a **fully automated, audit-ready pipeline** while maintaining **HIPAA-grade compliance**.
- Built a highly scalable **multi-database system** (PostgreSQL, MongoDB, Redis) supporting 15-30 microservices; engineered to support production traffic peaks of up to **500K requests** while balancing cross-store consistency with horizontal scalability.
- Developed **intelligent code generation pipelines** and **multi-modal requirement extraction tools**; built an LLM agent that analyzes existing repositories to identify ambiguous requirements before implementation, reducing prompt iterations by 40%.
- Evaluated and designed modular, scalable architectures for **intelligent automation platforms** (*Agentic Workflow*), translating complex UI/UX requirements into technical pipelines.
- **Technologies:** Python, FastAPI, LangGraph, OpenAI, Redis, MongoDB, Azure, Docker

Techolution

Generative AI Engineer Intern

Sep 2025 – Mar 2026

Hyderabad, India

- Designed the foundational architecture for a Pixel-to-Website AI tool; developed an optimized **Monte Carlo Tree Search (MCTS)** algorithm with parallel rollout execution for deterministic 1:1 UX-to-code translations.
- Managed a multi-tiered judging swarm utilizing actor-model concurrency, balancing inference cost and latency while navigating UI ambiguity with sub-second decision boundaries.
- Integrated a dynamic conflict-resolution pipeline for a Knowledge Retrieval agent; utilized vector quantization and asynchronous non-blocking I/O to proactively identify data gaps in enterprise-scale knowledge bases.
- **Technologies:** Python, PyTorch, Ray, Qdrant, Docker, AWS

NRSC - ISRO

Research Intern

Jun 2025 – Aug 2025

Hyderabad, India

- Evaluated and deployed **Vision Transformer (ViT)** architectures to denoise high-resolution SAR satellite imagery; leveraged hardware-accelerated tensor operations to enhance spatial resolution.
- Optimized deep learning inference pipelines utilizing **mixed-precision (FP16)** and batched execution, reducing VRAM footprint by 40% and accelerating throughput over baseline CNN models.
- **Technologies:** Python, PyTorch, CUDA, TensorRT, OpenCV

PROJECTS

High-Performance Distributed Key-Value Store | C++20, io_uring, Raft Consensus, Zero-Copy | [GitHub](#)

- Built a distributed key-value store from scratch, benchmarking at **1,640,631 Requests/sec** using **wrk** with Lua HTTP pipelining on a Dockerized Ubuntu 24.04 deployment (8 threads, 400 connections).
- Engineered a robust threading model to ensure deterministic request handling and high availability during extreme traffic spikes, verified through rigorous stress testing.
- Implemented the **Raft consensus algorithm** for strict fault tolerance and leader election, minimizing data loss through an append-only Write-Ahead Log (WAL) and Raft replication.

Low-Latency Distributed Message Broker | Java 22 (FFM API), io_uring, Netty, Disruptor | [GitHub](#)

- Created a distributed message broker measuring **1,000,000 ops/sec** on Linux by integrating **io_uring** via Java 22's Project Panama to streamline zero-copy network ingestion.
- Established comprehensive benchmarking and observability pipelines using JMH on an **arm64** architecture, systematically identifying and resolving JVM latency bottlenecks to achieve a **133.8ns** response time.
- Designed a robust binary wire protocol and resilient leader-follower replication system using **Netty**, ensuring strict data consistency and seamless failover across distributed nodes.

High-Frequency Multi-Agent Orchestration Engine | *Rust, Lock-Free Concurrency, Bare-Metal, Atomics* | [GitHub](#)

- Developed a highly resilient orchestration engine in **Rust** to manage massive agent swarms, focusing on predictable system behavior and strict resource allocation for mission-critical deployments.
- Designed reliable inter-agent communication buses, ensuring safe, concurrent state updates and preventing race conditions across thousands of parallel execution streams.
- Benchmarked a Multi-Consumer processing layer locally on Apple Silicon (8 CPU cores) utilizing NSW graph algorithms, dynamically partitioning hardware threads to achieve a peak throughput of **2.42 Million agents/sec**.

Distributed WebAssembly (WASM) Serverless Runtime | *Rust, Wasmtime, gRPC, JIT Compilation* | [GitHub](#)

- Implemented a distributed edge-compute runtime for deploying untrusted **WebAssembly** modules, engineering a secure provisioning pipeline that achieves **54.16µs cold-starts**.
- Engineered a fault-tolerant **Stateful Live Migration** protocol, guaranteeing seamless state continuity across physical network boundaries via **gRPC** during node failures or load balancing.
- Integrated the **Wasmtime JIT compiler** to enforce strict security boundaries and memory isolation, actively trapping out-of-bounds execution to ensure maximum system reliability against malicious payloads.

TECHNICAL SKILLS

Languages: Java, Python, C++, C, Rust, SQL, JavaScript/TypeScript

AI/ML: PyTorch, TensorFlow, Hugging Face, LangChain, LoRA/PEFT, RAG, FAISS, TensorRT, Generative AI

Systems & Backend: Distributed Systems, Java NIO, Linux io_uring, TCP Protocol Design, gRPC, REST APIs, FastAPI, Node.js

Databases & Data: PostgreSQL, MongoDB, Chroma DB, Redis, LevelDB, SQL query optimization

Cloud & DevOps: AWS (EC2, S3), Google Cloud Platform, Docker, Kubernetes, CI/CD

Core CS: Data Structures & Algorithms, System Design, Operating Systems, Computer Networks, Concurrency, OOP

ACHIEVEMENTS

- **Winner** - IBM Qiskit Fall Fest, 2024 (1/180)
- **Technical Excellence Award** - Silicon Labs Challenge, 2026 (4/440)
- **Best Startup Ideation Award** - ENTREPX, 2025 (1/90)